ALICE L WALTON SCHOOL OF MEDICINE

Acute Pericarditis

Last updated March 2, 2026 12 min read ScholarRx

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Editors: Jordan Finn, Russell Johnson

Adapted from: Acute Pericarditis

Listen to this Brick

0:00 / 19:35 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of acute pericarditis.
2. Explain the pathophysiology of acute pericarditis.
3. Interpret clinical findings to properly diagnose acute pericarditis.
4. Outline the management plan of acute pericarditis.

1).

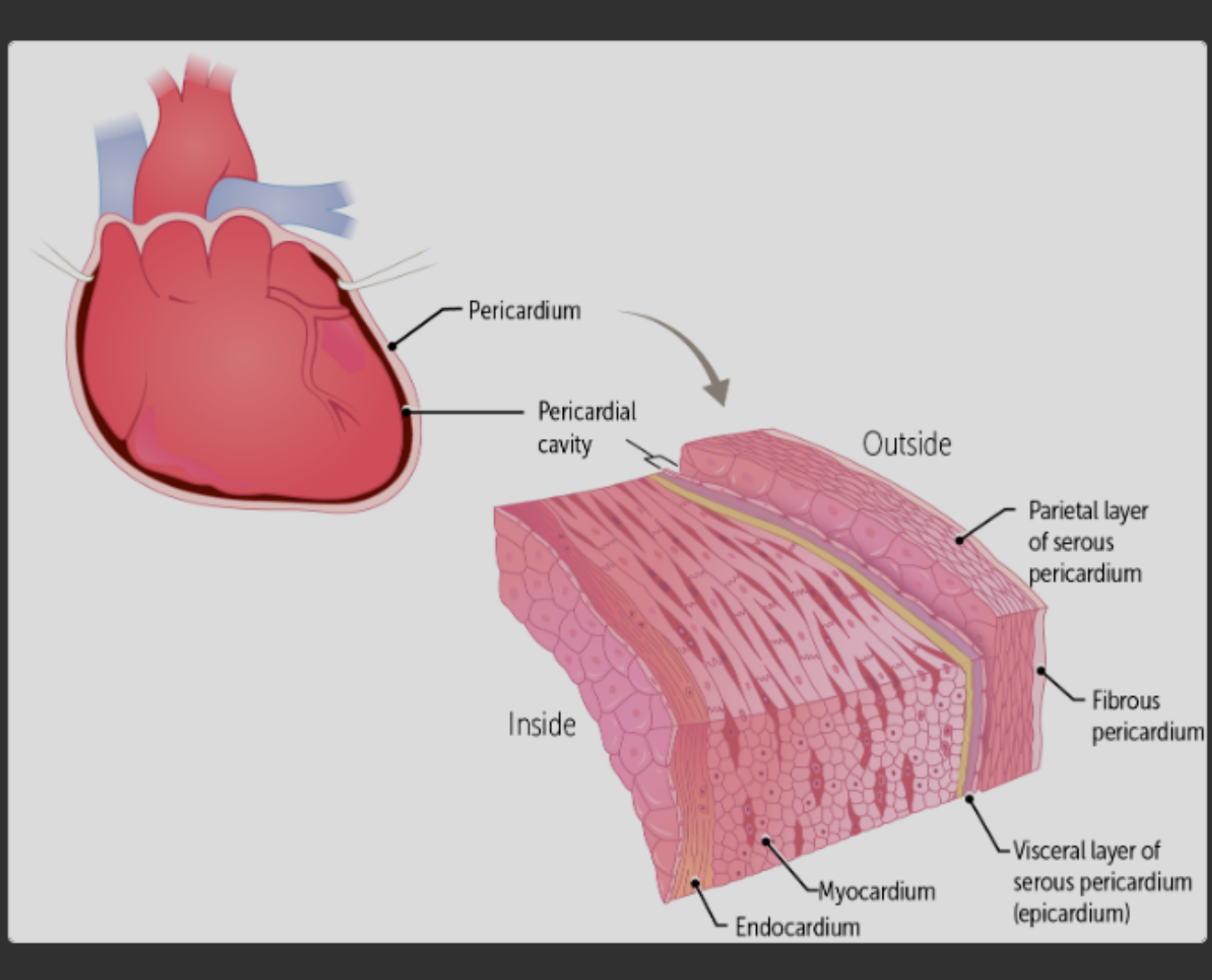
Pericardium
Pericardial cavity
Outside
Parietal layer of serous pericardium
Fibrous pericardium
Inside
Visceral layer of serous pericardium (epicardium)
Myocardium
Endocardium

Figure 1. Structure of the Normal Pericardium,

The pericardium is composed of three layers. From outer to inner, the layers are:

- Fibrous pericardium
- Parietal layer of serous pericardium
- Visceral layer of

In Figure 2, we can see a histologic view of normal visceral pericardium, which
amount of fluid can increase; this is known as a **pericardial effusion**.

What is the definition of acute pericarditis?

How Do Patients With Acute Pericarditis Present? (LO #1)

A patient with acute pericarditis classically presents with acute chest pain that is **pleuritic** in nature (i.e., sharp pain that is worsened by inspiration or coughing).

A classic presentation of the chest pain in acute pericarditis is that the pain improves when the patient is sitting up and leaning forward. Conversely, the chest pain worsens when lying flat. Why does this occur? The pain improves because sitting up and leaning forward reduces the pressure on the parietal layer of the serous pe nsing pain.

in the management of acute pericarditis.

Fibrinous Pericarditis. Acute fibrinous pericarditis is the most common type of acute pericarditis, and is characterized by the deposition of fibrin on the pericardial surfaces (Figure3).

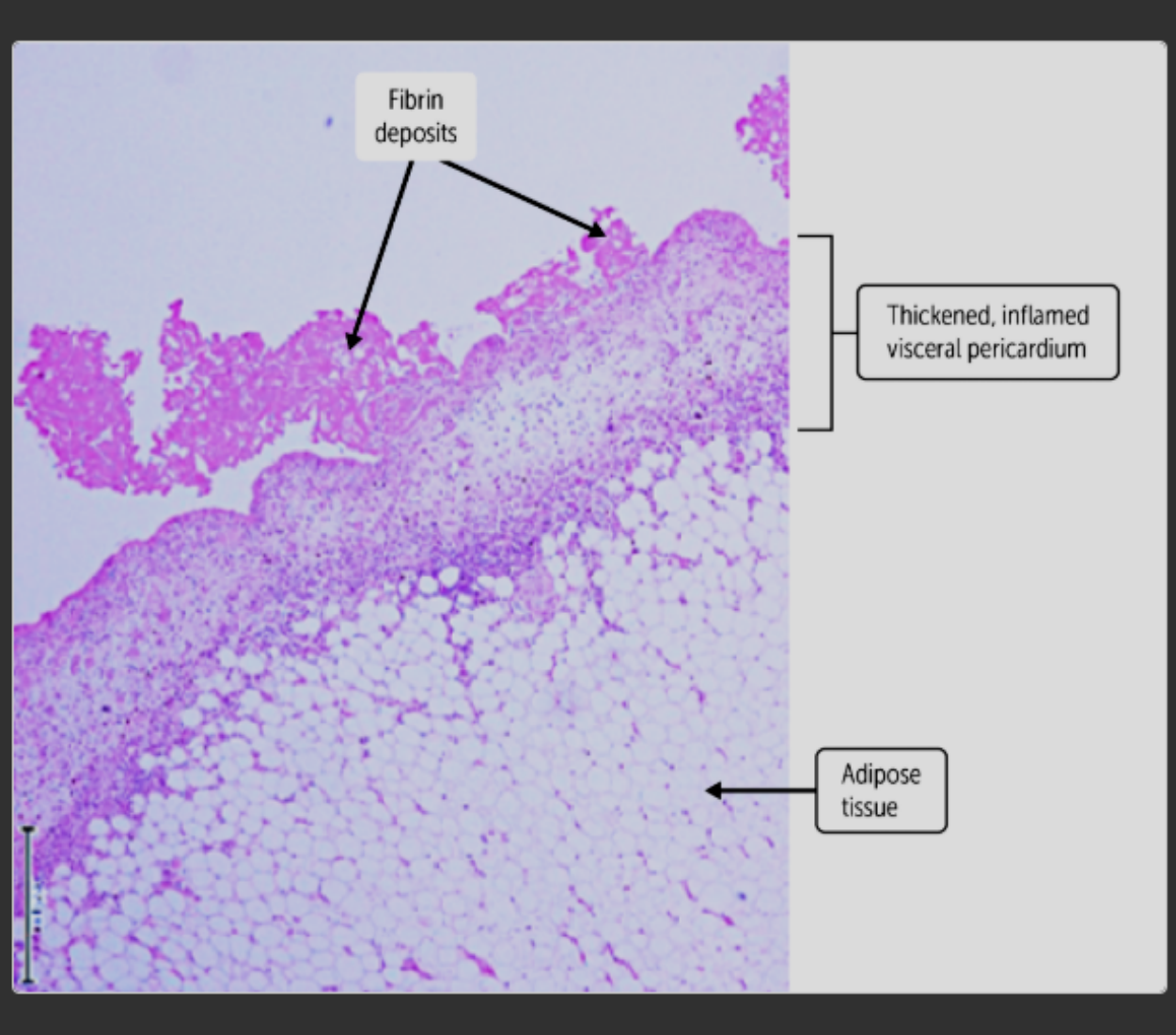

Figure 3. Histology of Acute Fibrinous Pericarditis.

Fibrinous pericarditis may be caused by:

- Myocardial infarc or late

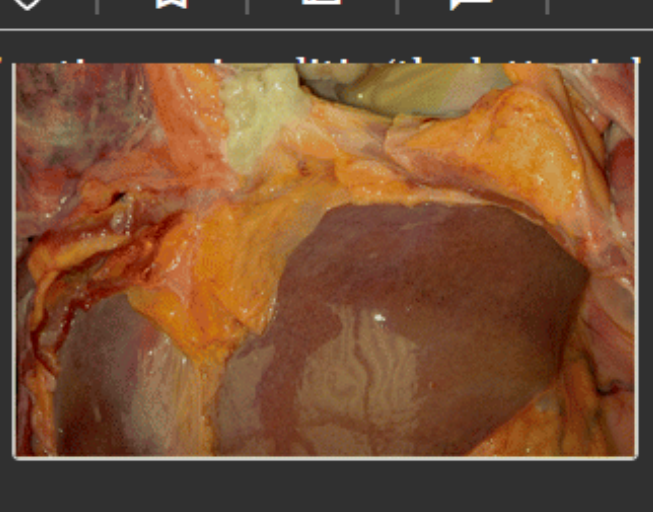

Figure 4. Purulent Pericarditis. Note the presence of pus within the pericardial cavity.

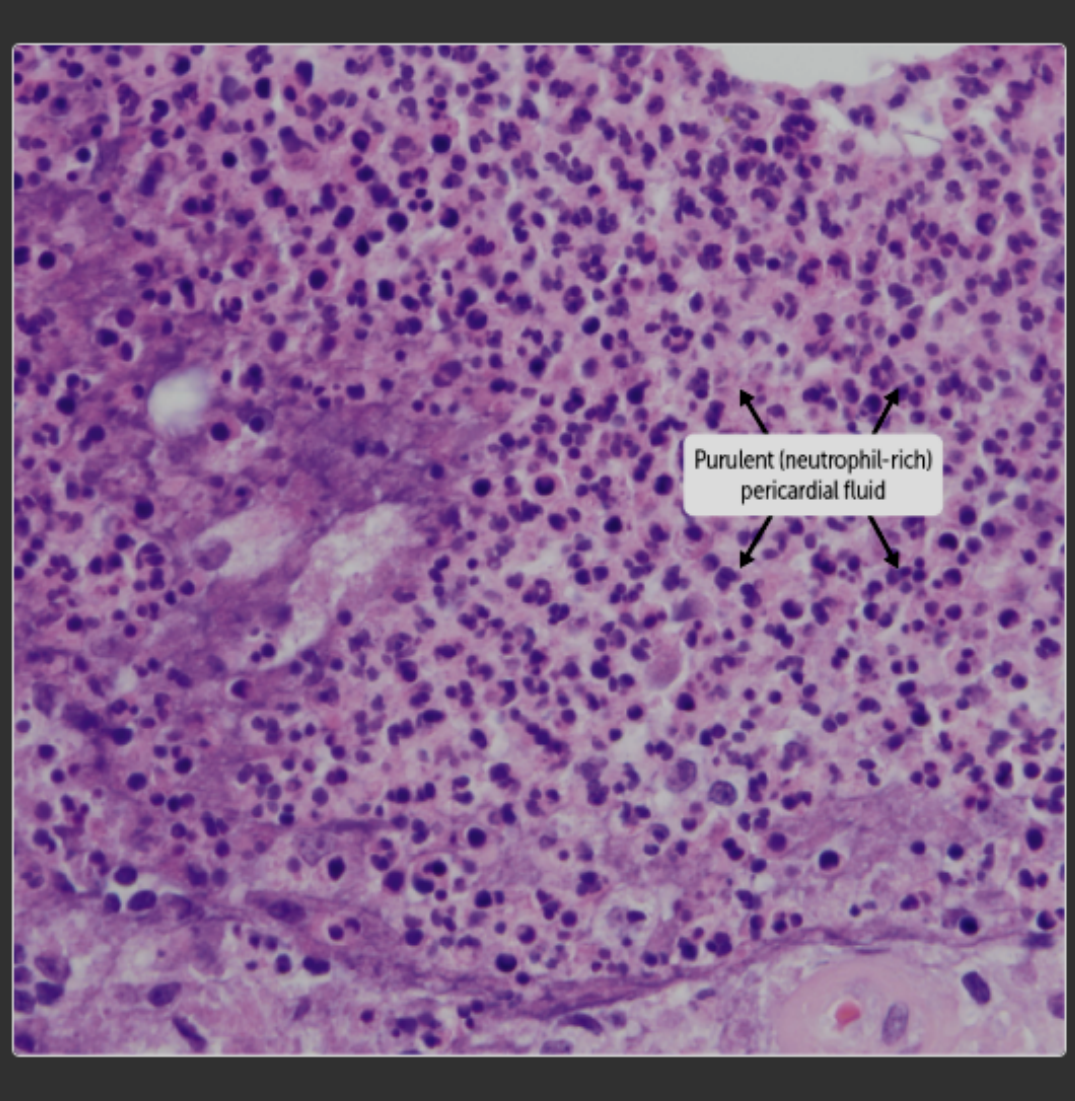

Figure 5. Histology of Acute Purulent Pericarditis. Purulent inflammation is characterized by a neutrophil-rich

How Do We Diagnose Acute Pericarditis? (LO #3)

The four principal diagnostic features of acute pericarditis are chest pain (sharp or pleuritic in nature), a pericardial friction rub, typical ECG findings, and (occasionally) a pleural effusion seen on echocardiogram. For most cases, further workup is usually not indicated unless the patient has suspected bacterial (purulent) pericarditis or cardiac tamponade.

Physical Examination

During the physical exam, a classic finding in acute pericarditis is a pericardial friction rub, though this may not always be present. It is a superficial “scratchy” sound produced when inflamed pericardial layers slide past each other with audible friction that can be detected via auscultation. A friction rub is not always present.

CLINICAL CORRELATION

A pericardial friction rub is a superficial “scratchy” sound produced when inflamed pericardial layers slide past each other with audible friction that can be detected via auscultation. The squeaky leather, sandpaper rubbing,

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Shock

Last updated March 8, 2026 21 min read ScholarRx

Adapted by: Phil Riley, Diptiman Bose, Ian Murray, Patrick Brooks

Editors: Russell Johnson, Jordan Finn

Adapted from: Shock

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Define shock.
2. Describe the clinical presentations of the different forms of shock.
3. Explain the physiologic parameters measured in shock.
4. Differentiate the five primary types of shock, including how they differ in card s.

What do you want to learn?

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pressures before succumbing to tissue hypoperfusion, whereas chronically hypertensive patients may experience the deleterious effects of hypoperfusion at higher-than-expected blood pressures. So, the assessment of shock requires both an objective measure of the vital signs and a more subjective assessment of organ perfusion.

What is a key indicator of shock despite normal blood pressure?

Pathophysiology of Shock

The cellular oxygen imbalance of shock is most commonly related to impaired oxygen delivery in the setting of circulatory failure. When cells cannot obtain sufficient oxygen, aerobic metabolism fails, forcing a shift to anaerobic glycolysis. This less efficient pathway produces lactate and generates inadequate amounts cell's primary energy currency.

- **Septic shock,** the most common form of distributive shock, results from a bacterial infection leading to life-threatening organ dysfunction. Microbial components, including lipopolysaccharides, stimulate inflammatory cells to produce cytokines such as transforming growth factor (TGF), interleukin-1 (IL-1), interferon γ, IL-6, IL-12, and IL-18. These cytokines mediate endothelial cell damage and the release of vasodilators such as nitric oxide and prostaglandin I_2 resulting in microcirculatory abnormalities. Septic shock patients often have fever, tachypnea, and a clear infectious source
- **Systemic inflammatory response syndrome (SIRS):** This is a complex inflammatory response to stress that involves the entire body. The presentations vary, and different patients may have tachycardia, hypothermia, hyperthermia, and peripheral WBC counts that are very low or very high. SIRS may lead to the failure of multiple organs (kidney, heart, and lungs). Causes can include sepsis but can also include noninfectious inflammation due to trauma, pancreatitis, or burns. There is activation of humoral and cellular immunity, the complement system, and cytokines.
- **Anaphylactic shock:** This is reexposure to an antigen against which an individual already has produced specific IgE antibodies. This leads to vasodilation through basophil degranulation and release of histamine, prostaglandins, and leukotrienes. Anaphylactic shock presents with urticaria, angioedema, bronchospasm, and a history of allergen exposure
- **Adrenal crisis:** This is a major deficiency of cortisol and/or aldosterone, often due to abrupt withdrawal of chronically administered corticosteroid drugs. These hormones are critical in maintaining the SVR, and there may be hypotension or shock when they are profoundly decreased. adrenal crisis may present emia, and hyperkalemia.

Hypovolemic Shock: Decreased Preload

Hypovolemic shock is caused by significant loss of blood or plasma volume, most often by a sudden **hemorrhage**. Far less often, it can result from routine causes of dehydration, like overuse of diuretics, severe vomiting or diarrhea, or lack of fluid intake. It can also result from the shifting of fluid from the vasculature to body cavities such as during pancreatitis, burns, or peritonitis. This is sometimes referred to as internal third-spacing.

In hypovolemic shock, less volume circulating in blood vessels means lower venous return, reduced ventricular filling, and reduced preload. The reduced preload leads to reduced stroke volume, which reduces cardiac output. SVR increases as a compensatory mechanism to maintain blood pressure via arteriolar constriction.

The body compensates for hypovolemia through multiple mechanisms: baroreceptor-mediated sympathetic activation increases heart rate and contractility while causing vasoconstriction in non-essential vascular beds (skin, splanchnic circulation, kidneys); the renin-angiotensin-aldosterone system promotes sodium and water retention; and antidiuretic hormone release further enhan nisms can temporarily maintain volume loss, masking